

Model	Architecture Method	/	KL Divergence (Frequencies)	Co-occurrence MSE (Locality)	Features and Conclusions
Baseline	Random by frequencies		0.0178	2.25e-04	Ideal global proportions, but complete chaos (noise) on the map. Lack of biome logic.
Naive Bayes	Pixel-wise statistics		0.0003	1.99e-07	Fantastic compliance with neighborhood rules, but generates infinite straight lines (patterns).
GMM	EM algorithm + PCA (Latent space)		0.0203	3.01e-06	Perfectly captures the outlines of global continents, but produces blurred block boundaries.
MRF	Gibbs sampling		0.0005	1e-07	Strictly follows neighborhood physics, perfectly smooths lines, but works slowly.
Ensemble	GMM (Global) + MRF (Local)		~ 0.001	$\sim 1e-07$	Best statistical result. GMM draws continents, and MRF finishes drawing perfect sand and flowers.
GAN512	Neural Network		1.3736	3.39e-04	Numbers are worse, as the AI "improvises" and changes the map balance, but visually this is the most creative result.

Table 1: Model Comparison